ShopperHub

Software Requirements Document

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Project: Full-Stack E-Commerce Platform

Repository: perfactprachi-beep/ShopperHub

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1. Project Overview

1.1 Project Name

ShopperHub (internally: ShoppersStop Clone)

1.2 Purpose

ShopperHub is a full-stack, multi-category fashion and lifestyle e-commerce platform enabling customers to browse products, manage carts, place orders via online payment or cash-on-delivery, track deliveries, and redeem loyalty points. Administrators manage inventory, orders, banners, coupons, CMS pages, and store operations through a dedicated admin panel.

1.3 Tech Stack

Layer	Technology
Frontend	React 18, Vite 5, Tailwind CSS, Zustand, React Router v6
Backend	Node.js (ESM), Express 5
Database	PostgreSQL (via `pg` pool)
Payments	Razorpay
SMS	Fast2SMS (India), Twilio (international)
Auth	JWT (access + refresh tokens), bcrypt
File Storage	Local disk via Multer + Sharp (image processing)
PWA	Vite PWA Plugin + Workbox
Security	Helmet, CORS, express-rate-limit

1.4 Architecture Overview

Layered monolith with clear separation: React SPA → Vite proxy → Express REST API → PostgreSQL. Frontend state managed via Zustand stores. Backend follows Routes → Controllers → Services/Queries pattern.

2. Functional Requirements

Authentication

ID	Description	Input	Output	Business Logic
FR-001	User Registration	full_name, email, phone, password	JWT tokens + user	bcrypt hash; issue access + refresh tokens; httpOnly cookie
FR-002	User Login	email, password	JWT tokens + user	Verify hash; issue tokens; rate-limited 10 req/min
FR-003	Token Refresh	Refresh token (cookie)	New access token	Verify refresh token; issue new access token
FR-004	User Logout	Auth header	Success	Invalidate session
FR-005	OTP Verification	phone, OTP	Verified status	Fast2SMS (India) or Twilio (international)
FR-006	Admin Login	email, password	JWT with role=admin	Validates role='admin'

Product Catalogue

ID	Description	Input	Output	Business Logic
FR-007	List Products	category, brand, gender, price, discount, sort, page	Paginated list	Recursive CTE for subcategory inclusion
FR-008	Product Detail	slug	Product + images + variants + attributes	Parallel queries for related data
FR-009	Search Products	query string	Ranked list	PostgreSQL full-text search + ILIKE fallback
FR-010	Filter Products	price range, discount, brand, category, gender	Filtered list	Dynamic parameterized WHERE clause
FR-011	Sort Products	sort param	Sorted list	price_asc, price_desc, newest, discount, random
FR-012	Product Variants	product_id	sizes, colors, stock	Each variant has independent stock

Cart

ID	Description	Input	Output	Business Logic
FR-013	Add to Cart	productId, variantId, quantity	Updated cart	Upsert — increment if exists; validate stock
FR-014	View Cart	userId	Items with line totals	Join with product/variant/image data

FR-015	Update Quantity	cartItemId, quantity	Updated item	Validate stock before update
FR-016	Remove Item	cartItemId	Success	Hard delete from cart_items
FR-017	Clear Cart	userId	Success	Auto-triggered post-order fulfilment
FR-018	Guest Cart Merge	guestItems array	Merged cart	On login, merge local storage cart into DB

Checkout & Orders

ID	Description	Input	Output	Business Logic
FR-022	Create Razorpay Order	addressId, couponCode, usePoints, deliveryType	Razorpay orderId + amount	Batch stock check; coupon validation; points cap 20%
FR-023	Verify Payment	orderId, paymentId, signature	orderId	HMAC-SHA256 verification; atomic DB transaction
FR-024	Create COD Order	addressId, couponCode, usePoints, deliveryType	orderId	Skips Razorpay; payment_status='pending'
FR-025	Shipping Calculation	deliveryType, subtotal	Shipping amount	Standard: free ≥ ₹999 else ₹99; Express: ₹199; Pickup: ₹0
FR-026	Order Fulfilment	order data + items	Confirmed order	BEGIN → insert order → insert items → lock+deduct stock → award points → mark coupon → clear cart → COMMIT
FR-027	Cancel Order	orderId	Cancelled order	Only pending/confirmed; restocks variants; deducts earned points
FR-028	Order History	userId, page, limit	Paginated orders	Aggregated JSON items per order
FR-029	Order Detail	orderId	Order + items + address	Validates ownership
FR-030	Order Tracking	orderId	Status + expected delivery	Returns delivery_type, status, expected_by

Loyalty Points, Coupons & Addresses

ID	Description	Business Logic
FR-031	Earn Points	1 point per ₹100 spent; awarded in fulfilment transaction
FR-032	Redeem Points	Max 20% of post-coupon subtotal; 1 point = ₹1
FR-033	Points Balance	Returned in user profile
FR-034	Apply Coupon	Validates: active, not expired, min_order met, per-user + global usage limits
FR-036	Add Address	Max per user enforced; ownership validated
FR-038	Delete Address	Soft delete (is_deleted flag) — never hard-deleted

Delivery

ID	Description	Business Logic
FR-040	Express Delivery Check	Pincode lookup in express_delivery_pincodes table
FR-041	Store Pickup	Active stores with address + timing
FR-042	Store Pickup PIN	4-digit PIN generated at order creation

Admin Panel

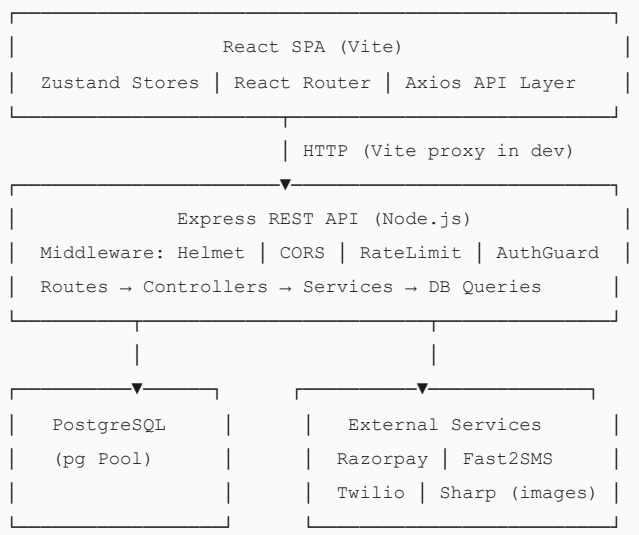
ID	Description	Business Logic
FR-049	Dashboard Stats	Parallel DB queries; 30-day sales trend
FR-050	Product CRUD	Create/update/soft-delete; image upload via Multer + Sharp
FR-051	Variant Management	Per-product; independent stock per variant
FR-052	Category CRUD	Hierarchical (parent/child)
FR-053–055	Brand / Banner / Coupon CRUD	Full CRUD with validation
FR-057	Order Management	Status transitions trigger in-app notifications
FR-058	User Management	Cannot modify/delete admin accounts
FR-059	Inventory Dashboard	Warehouse-aware; low stock alerts
FR-061	CMS Pages	Admin-managed content at /pages/:slug
FR-062–063	Store & Pincode Management	CRUD for pickup stores and express pincodes

3. Non-Functional Requirements

ID	Category	Requirement
NFR-001	Security	All passwords hashed with bcrypt (min 10 rounds)
NFR-002	Security	JWT access tokens short-lived; refresh tokens in httpOnly cookies
NFR-003	Security	Helmet CSP: only Razorpay, Google Fonts, and self allowed
NFR-004	Security	Auth endpoints rate-limited to 10 requests/minute per IP
NFR-005	Security	Global rate limit: 1000 requests per 15 minutes per IP
NFR-006	Security	Payment signature verified server-side via HMAC-SHA256
NFR-007	Security	Admin routes protected by `authGuard` + `adminGuard` (role check)
NFR-008	Performance	All API responses gzip-compressed via compression middleware
NFR-009	Performance	DB connection pooling via pg.Pool — no per-request connections
NFR-010	Performance	Stock pre-check uses single batch ANY($1) query — no N+1
NFR-011	Performance	Product images processed/resized via Sharp before storage
NFR-012	Reliability	DB startup retries up to 5 times with exponential backoff
NFR-013	Reliability	Order fulfilment wrapped in DB transaction — atomic, rollback on failure
NFR-014	Reliability	unhandledRejection + uncaughtException handlers prevent silent crashes
NFR-015	Scalability	Stateless JWT auth — horizontally scalable (no server-side sessions)
NFR-016	Scalability	Config fully externalised via .env — 12-factor compliant
NFR-017	Scalability	Health (/api/health) and readiness (/api/ready) endpoints for orchestration
NFR-018	Maintainability	Layered architecture: routes → controllers → services → queries
NFR-019	Maintainability	ESM modules throughout — no CommonJS mixing
NFR-020	Availability	PWA with Workbox offline caching for static assets and Google Fonts
NFR-021	Testing	No automated tests present — critical gap before production

4. System Architecture

4.1 Component Diagram



4.2 Checkout Data Flow

```
User selects address/delivery
      → POST /api/payments/create-order
      → Batch stock check (ANY query)
      → Coupon validate
      → Points calc
      → Razorpay order created
      → Frontend opens Razorpay modal
      → Payment success callback
      → POST /api/payments/verify
      → HMAC-SHA256 verify
      → fulfillOrder (DB transaction: BEGIN…COMMIT)
      → SMS confirmation (fire-and-forget)
      → Response with orderId
```

4.3 External Integrations

Service	Purpose	Config Keys
Razorpay	Online payment processing	RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET
Fast2SMS	SMS for Indian numbers (+91)	FAST2SMS_API_KEY
Twilio	SMS for international numbers	TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_PHONE
Google Fonts	Typography (CSP allowlisted)	CDN, no key required

5. Data Requirements

5.1 Core Entities

Entity	Key Fields	Relationships
`users`	id, full_name, email, phone, password_hash, role, first_citizen_points, is_active	has many orders, addresses, cart_items, wishlists, reviews
`products`	id, title, slug, brand_id, category_id, base_price, discount_pct, gender, stock, status, is_deal, is_returnable	belongs to brand, category; has many variants, images, attributes
`product_variants`	id, product_id, size, color, sku, stock, extra_price	belongs to product; referenced in cart, order_items, inventory
`categories`	id, name, slug, parent_id	self-referential (parent/child); recursive CTE for querying
`orders`	id, user_id, address_id, coupon_id, subtotal, discount, shipping, total, points_earned, payment_method, payment_status, status, delivery_type, delivery_method, store_id, pickup_pin, expected_by	belongs to user, address; has many order_items
`order_items`	id, order_id, product_id, variant_id, quantity, unit_price	belongs to order, product, variant
`addresses`	id, user_id, label, line1, city, state, pincode, is_deleted	belongs to user; soft-deletable
`coupons`	id, code, discount_type, discount_value, min_order, max_uses, used_count, per_user_limit, expires_at, is_active	referenced in orders
`inventory`	id, variant_id, warehouse_id, stock_quantity	mirrors product_variants stock
`inventory_logs`	id, inventory_id, action_type, quantity, notes, created_at	audit trail for stock changes
`stores`	id, name, city, address, pincode, timing, is_active	pickup locations
`notifications`	id, user_id, message, read_at, created_at	belongs to user
`pages`	id, title, slug, content, is_active	CMS pages

5.2 Key Constraints

Field	Constraint
products.slug	UNIQUE
users.email	UNIQUE
coupons.code	UNIQUE
orders.status	ENUM: pending, confirmed, shipped, delivered, cancelled
orders.payment_status	ENUM: pending, paid, failed
orders.delivery_type	ENUM: standard, express, store_pickup

product_variants.stock	Non-negative (GREATEST constraint on deduction)
users.first_citizen_points	Non-negative (GREATEST constraint)
addresses.is_deleted	Soft delete — never hard-deleted

6. User Roles & Permissions

6.1 Roles

Role	Description
Guest	Unauthenticated visitor
Customer	Registered, authenticated user
Admin	Platform administrator (role = 'admin' in DB)

6.2 Access Matrix

Feature	Guest	Customer	Admin
Browse products/categories	✅ Yes	✅ Yes	✅ Yes
Search products	✅ Yes	✅ Yes	✅ Yes
Add to cart	Local only	✅ DB	❌ No
Add to wishlist	❌ No	✅ Yes	❌ No
Checkout / Place order	❌ No	✅ Yes	❌ No
View order history	❌ No	Own only	All orders
Cancel order	❌ No	Own (pending/confirmed)	✅ Yes
Submit review	❌ No	Purchased only	❌ No
Admin dashboard	❌ No	❌ No	✅ Yes
Product / Variant CRUD	❌ No	❌ No	✅ Yes
Order status update	❌ No	❌ No	✅ Yes
User management	❌ No	❌ No	✅ Yes
Inventory management	❌ No	❌ No	✅ Yes
CMS page management	❌ No	❌ No	✅ Yes
Payment method config	❌ No	❌ No	✅ Yes

7. API / Interface Requirements

7.1 Base URL & Response Envelope

Base URL (dev): `http://localhost:5000/api`

All responses: `{ success: boolean, data?: any, message?: string }`

7.2 Auth Endpoints

Method	Endpoint	Auth	Description
POST	/auth/register	None	Register user
POST	/auth/login	None	Login
POST	/auth/refresh	Cookie	Refresh access token
POST	/auth/logout	Bearer	Logout
POST	/auth/send-otp	None	Send OTP via SMS
POST	/auth/verify-otp	None	Verify OTP

7.3 Product Endpoints

Method	Endpoint	Auth	Description
GET	/products	None	List with filters/sort/pagination
GET	/products/:slug	None	Product detail
GET	/products/search?q=	None	Full-text search

7.4 Cart Endpoints

Method	Endpoint	Auth	Description
GET	/cart	Bearer	View cart
POST	/cart	Bearer	Add item
PUT	/cart/:id	Bearer	Update quantity
DELETE	/cart/:id	Bearer	Remove item
POST	/cart/merge	Bearer	Merge guest cart on login

7.5 Payment Endpoints

Method	Endpoint	Auth	Description
GET	/payments/methods	None	List active payment methods
POST	/payments/create-order	Bearer	Create Razorpay order

POST	/payments/create-cod-order	Bearer	Place COD order
POST	/payments/verify	Bearer	Verify payment + fulfil order
GET	/payments/orders/:id	Bearer	Fetch order by ID

7.6 Admin Endpoints (Bearer + admin role required)

Method	Endpoint	Description
GET	/admin/dashboard/stats	Dashboard metrics
GET/POST/PUT/DELETE	/admin/products	Product CRUD
GET/POST/PUT/DELETE	/admin/categories	Category CRUD
GET/POST/PUT/DELETE	/admin/brands	Brand CRUD
GET/POST/PUT/DELETE	/admin/banners	Banner CRUD
GET/POST/PUT/DELETE	/admin/coupons	Coupon CRUD
GET/POST/PUT/DELETE	/admin/offers	Offer CRUD
GET	/admin/orders	List all orders
PUT	/admin/orders/:id/status	Update order status
GET/PATCH/DELETE	/admin/users	User management
GET/POST/PUT/DELETE	/admin/payment-methods	Payment config
GET/POST/PATCH/DELETE	/admin/delivery/stores	Store management
GET/POST/PATCH/DELETE	/admin/delivery/pincodes	Pincode config
GET/POST/PUT/DELETE	/admin/pages	CMS pages

7.7 Standard HTTP Status Codes

Code	Meaning
200	Success
201	Created
400	Bad request / validation failure
401	Unauthenticated
403	Forbidden (wrong role)
404	Resource not found
409	Conflict (out of stock, duplicate)
503	Service unavailable (DB down)

8. Assumptions & Constraints

8.1 Assumptions

ID	Assumption
A1	Currency is Indian Rupees (INR) exclusively; Razorpay amounts are in paise (×100)
A2	Phone numbers with +91 prefix use Fast2SMS; all others use Twilio
A3	One user can have multiple saved addresses; all are soft-deleted, never hard-deleted
A4	A product review requires a delivered order containing that product
A5	Loyalty points have no expiry — permanent until redeemed or order cancelled
A6	Express delivery availability is pincode-gated via database config
A7	Store pickup generates a 4-digit PIN at order creation
A8	Image uploads are stored on local disk (/uploads) — no cloud storage configured

8.2 Constraints

ID	Constraint
C1	No automated test suite — all testing is currently manual
C2	No API versioning (/api/v1/) — breaking changes affect all clients immediately
C3	Local file storage does not scale horizontally — S3/CDN migration needed for multi-server deployment
C4	No refresh token revocation store — logout does not truly invalidate tokens until expiry
C5	Migration scripts are numbered manually with no version tracking tool
C6	No structured logging — all output via console.log/error; no log aggregation support
C7	Razorpay is the only online payment gateway; no PayPal, UPI QR, or wallet support

9. Glossary

Term	Definition
First Citizen Points	Loyalty reward points earned at 1 point per ₹100 spent; redeemable at checkout (1 point = ₹1, max 20% of order)
Variant	A product SKU with specific size/color combination and its own stock count
Fulfilment	The atomic DB transaction that creates an order, deducts stock, awards points, marks coupon used, and clears cart
Razorpay Order	A pre-payment object created on Razorpay's servers representing the intended transaction amount
HMAC Verification	Server-side payment signature check using SHA-256 to confirm Razorpay payment authenticity
Store Pickup	Delivery option where customer collects from a physical store using a PIN
Express Delivery	Pincode-gated premium delivery at ₹199 flat; availability stored in express_delivery_pincodes table
Soft Delete	Marking a record as inactive (is_deleted=true) instead of removing it from the database
authGuard	Express middleware that validates JWT Bearer token on protected routes
adminGuard	Express middleware that enforces role='admin' on administrative routes
CMS Page	Admin-managed static content page rendered at /pages/:slug on the frontend
Inventory Log	Audit trail entry recording stock changes (order deduction, return restock, manual adjustment)
lineTotal	Cart item total = unitPrice × quantity (computed in DB query)
paise	Indian sub-unit of currency (1 INR = 100 paise); Razorpay requires amounts in paise
Pickup PIN	4-digit random code generated at store-pickup order creation

ShopperHub

Master Technical Document

Software Requirements · Architecture Review · Workflow Documentation

Version: 2.0 | **Date:** May 2026 | **Status:** Draft — Pre-Production

Repository: perfactprachi-beep/ShopperHub | **Branch:** main

Prepared by: Principal Software Architect | **Classification:** Confidential

Stack: React 18 · Node.js/Express 5 · PostgreSQL · Razorpay · Fast2SMS · Twilio

Part 1 · SRD | Part 2 · Architecture Review | Part 3 · Workflow Docs | 68 FR · 21 NFR · 10 Phases

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PART 2 — ARCHITECTURE & CODE REVIEW

PART 3 — COMPLETE E-COMMERCE WORKFLOW

PART 1

Software Requirements Document

Complete functional and non-functional requirements derived from source code analysis. Covers all modules including product management, inventory, cart, orders, payments, loyalty points, and admin operations.

1.1 Project Overview

Project Name & Purpose

ShopperHub is a full-stack, multi-category fashion and lifestyle e-commerce platform modelled after ShoppersStop. It enables customers to browse products across categories (Men, Women, Kids, Beauty, Luxe, Home), manage carts and wishlists, checkout via Razorpay online payment or Cash on Delivery, track orders, and earn/redeem loyalty points. Administrators manage the full product catalogue, inventory, orders, banners, coupons, CMS content, delivery configuration, and store pickup operations through a dedicated admin panel.

Tech Stack

Layer	Technology	Version	Notes
Frontend	React + Vite	18.3 / 5.4	SPA, lazy-loaded routes, PWA-enabled
State Management	Zustand	4.5	Auth, cart, wishlist, toast stores
Styling	Tailwind CSS	3.4	Utility-first; custom config
Routing	React Router	v6.23	Nested routes, ProtectedRoute guard
Backend	Node.js + Express	22 / 5.0	ESM modules throughout
Database	PostgreSQL	Latest	Raw SQL via pg.Pool; port 5433
Auth	JWT + bcrypt	9.0 / 5.1	Access + refresh tokens; httpOnly cookies
Payments	Razorpay	2.9	Order create + HMAC verification
SMS	Fast2SMS + Twilio	—	India (+91) vs international routing
File Processing	Multer + Sharp	—	Upload + resize images; local /uploads
Security	Helmet + cors + rate-limit	7.1	CSP, CORS, 1000 req/15min global limit
PWA	Vite PWA + Workbox	—	Offline caching for assets & fonts

Architecture Pattern

Layered Monolith — Frontend SPA served separately from backend API. Backend follows strict layering: `Routes → Controllers → Services/Queries → Database`. No microservices; single PostgreSQL instance. Horizontally scalable via stateless JWT auth, but local file storage is a current scaling blocker.

1.2 Functional Requirements

Authentication Module

ID	Description	Input	Output	Business Logic
FR-001	User Registration	full_name, email, phone, password	JWT access token + user object	bcrypt hash (10 rounds); issue access + refresh JWT; set httpOnly cookie
FR-002	User Login	email, password	JWT tokens + user	Verify bcrypt hash; rate-limited 10 req/min via authLimiter
FR-003	Token Refresh	Refresh token (httpOnly cookie)	New access token	Verify refresh JWT secret; issue new short-lived access token
FR-004	Logout	Bearer token	Success message	Clear cookie; note: no server-side revocation store
FR-005	OTP Send	phone number	OTP sent via SMS	Route: +91 → Fast2SMS; other → Twilio; 6-digit OTP
FR-006	OTP Verify	phone, OTP	Verified status	Time-limited OTP validation
FR-007	Admin Login	email, password	JWT with role=admin	Same flow as FR-002; adminGuard enforces role check on all /api/admin/* routes

Product Management Module

ID	Description	Input	Output	Business Logic
FR-008	List Products (storefront)	category, brand, gender, minPrice, maxPrice, minDiscount, sort, page, limit	Paginated product list with total_count	Recursive CTE for subcategory inclusion; only status='active' products shown
FR-009	Product Detail	slug	Product + images + variants + attributes	Three parallel queries; only active products returned
FR-010	Search Products	q (query string)	Ranked product list	PostgreSQL full-text `tsvector` + `ts_rank`; ILIKE fallback on brand/category name
FR-011	Sort Products	sort param	Ordered list	Enum: price_asc, price_desc, newest, discount, random (RANDOM())
FR-012	Admin: Create Product	title, slug, brand_id, category_id, description, gender, base_price, discount_pct, stock, status, is_deal, is_returnable + images	Created product	Images uploaded via Multer; processed by Sharp; stored in /uploads; first image set as primary
FR-013	Admin: Update Product	id + product fields	Updated product	Full field replacement; returns 404 if not found
FR-014	Admin: Soft-Delete Product	id	Success	Sets status='inactive'; product removed from storefront immediately
FR-015	Admin: Bulk Image Fill	— (none)	Count of filled products	Finds active products with no images; assigns placeholder via `getProductPlaceholder()`

Variant & Inventory Management Module

ID	Description	Input	Output	Business Logic
FR-016	Add Product Variant	product_id, size, color, sku, stock, extra_price	Created variant	Each variant has independent stock; no auto-creation of inventory row

FR-017	Update Variant	variant_id + fields	Updated variant	Full replacement of size, color, sku, stock, extra_price
FR-018	Delete Variant	variant_id	Success	Hard delete; cascades via FK constraints
FR-019	Set Variant Stock (Inventory Module)	variantId, stock_quantity, low_stock_threshold, notes	Inventory record + updated variant	`setVariantStock()` : upserts inventory row; updates product_variants.stock atomically; creates inventory log
FR-020	Adjust Stock (delta)	inventory_id, quantity_change, action_type, notes	Updated inventory	Adds/subtracts from current stock; logs adjustment; syncs product_variants
FR-021	Bulk Stock Update	updates[] array of {id, stock_quantity}	Updated records	Sequential updates; no transaction wrapper — partial failure leaves inconsistent state
FR-022	Low Stock Alert List	threshold, page, limit	Variants below threshold	Compares inventory.stock_quantity < low_stock_threshold
FR-023	Warehouse CRUD	name, location, manager_name, contact_number	Warehouse record	Soft-delete (is_active=false); inventory items remain orphaned
FR-024	Inventory Dashboard Stats	—	total_products, total_warehouses, in_stock, low_stock, out_of_stock counts	Parallel queries across warehouse + inventory + product_variants tables
FR-025	Inventory Logs	page, limit, filters	Paginated audit log	Types: order_deduction, return_restock, manual_adjustment, variant_stock_set

Cart Management Module

ID	Description	Input	Output	Business Logic
FR-026	View Cart	userId (from JWT)	items[], subtotal, itemCount	JOIN product/variant/image; `buildResponse()` computes lineTotal server-side
FR-027	Add to Cart	productId or variantId, quantity	Updated cart	If no variantId: find/create default variant (null size/color); stock check before add; UPSERT on conflict
FR-028	Update Cart Quantity	itemId, quantity	Updated cart	quantity must be ≥ 1; ownership validated (user_id match)
FR-029	Remove Cart Item	itemId	Updated cart	Hard delete; ownership validated
FR-030	Clear Cart	userId	Empty cart	Auto-triggered inside `fulfillOrder()` transaction after successful order
FR-031	Guest Cart Merge	guestItems[] array (variantId, quantity)	Merged DB cart	On login: client sends localStorage cart items; server upserts each into DB cart

Order Lifecycle Module

ID	Description	Input	Output	Business Logic
FR-032	Create Razorpay Order	addressId, couponCode, usePoints, deliveryType	razorpayOrderId, amount, keyId, subtotal, discount, shipping, total	Batch stock pre-check (ANY query); coupon validate; points cap 20%; `calcShipping()` ; Razorpay API call
FR-033	Verify Payment & Fulfill	razorpayOrderId, razorpayPaymentId, razorpaySignature, addressId, couponId, totals, deliveryType	orderId + success message	HMAC-SHA256 verify; calls `fulfillOrder()` atomic transaction; SMS confirmation
FR-034	Place COD Order	addressId, couponCode, usePoints, deliveryType, storeId	orderId + success message	Same as FR-032 without Razorpay; payment_status='pending'; direct fulfillment

FR-035	Order Fulfillment (atomic)	Full order data + items[]	Confirmed order record	BEGIN → INSERT orders → INSERT order_items → FOR UPDATE lock variants → validate stock ≥ qty → UPDATE stock → sync inventory → award points → mark coupon used → DELETE cart → COMMIT; ROLLBACK on any failure
FR-036	Cancel Order	orderId, userId	Cancelled order	Only status IN ('pending','confirmed'); transaction: set cancelled → restore variant stock → restore inventory → log return_restock → deduct earned points; COMMIT
FR-037	View Order History	userId, page, limit	Paginated orders with items (JSON_AGG)	Grouped by order; items aggregated per order in single query
FR-038	Order Detail	orderId	Order + items + address + store info	Ownership check (user_id = $2); includes store timing for pickup orders
FR-039	Admin: List All Orders	page, limit, status filter	Paginated orders with customer info	Joined with users; total_count via window function
FR-040	Admin: Update Order Status	orderId, status	Updated order + notification	Status transitions: confirmed → shipped → delivered; fires in-app notification

Payment Integration Module

ID	Description	Input	Output	Business Logic
FR-041	List Payment Methods	—	Active payment methods	Public endpoint; admin-configurable via `payment_methods` table
FR-042	Razorpay Order Create	amount (₹), currency, receipt	Razorpay order object	`Math.round(amount * 100)` converts to paise; `payment_capture: 1` for auto-capture
FR-043	Payment Signature Verify	orderId, paymentId, signature	Boolean valid/invalid	HMAC-SHA256: `orderId|paymentId` signed with RAZORPAY_KEY_SECRET; constant-time comparison
FR-044	Admin: Configure Payment Methods	method, is_active, display config	Updated config	Toggle Razorpay / COD availability for storefront

Coupon & Discount Engine

ID	Description	Input	Output	Business Logic
FR-045	Validate & Apply Coupon	couponCode, subtotal, userId	{valid, discount, couponId, reason}	Chain: is_active → not expired → min_order met → per_user_limit not exceeded → global max_uses not exceeded → compute discount (flat or %)
FR-046	List Active Coupons	—	Public coupon list	Returns is_active + not expired coupons for storefront display
FR-047	Admin: Coupon CRUD	code, discount_type, discount_value, min_order, max_uses, per_user_limit, expires_at	Coupon record	Full CRUD; used_count tracked; unique code constraint
FR-048	Mark Coupon Used	couponId	Incremented used_count	Called inside `fulfillOrder()` transaction; atomically committed with order

Loyalty Points (First Citizen Points)

ID	Description	Input	Output	Business Logic
FR-049	Earn Points on Order	order total	Points credited to user	1 point per ₹100 (floor); awarded inside `fulfillOrder()` transaction atomically
FR-050	Redeem Points at Checkout	usePoints flag, current points balance, subtotal	pointsDiscount amount	Cap: MIN(user_points, FLOOR((subtotal - coupon_discount) × 0.20)); 1 point = ₹1
FR-051	Deduct Points on Cancel	orderId, points_earned	Reduced balance	GREATEST(points - earned, 0) inside cancel transaction; prevents negative balance

FR-052	Points Balance	userId	first_citizen_points	Returned in user profile; non-negative guaranteed by DB constraint

Storefront / Homepage & Delivery

ID	Description	Input	Output	Business Logic
FR-053	Homepage Sections	—	Banners, featured products, categories, offers	Parallel DB queries; only active records returned
FR-054	Shipping Calculation	deliveryType, subtotal	Shipping amount (₹)	Standard: ₹0 if subtotal ≥ ₹999 else ₹99; Express: ₹199 flat; Store Pickup: ₹0
FR-055	Express Delivery Check	pincode	is_express, delivery_hrs	Lookup in `express_delivery_pincodes` table; returns availability flag
FR-056	Store Pickup	—	Active stores with address + timing	is_active=true stores only; returns timing for customer display
FR-057	Store Pickup PIN	orderId (store_pickup orders)	4-digit PIN	Generated at order creation: `Math.floor(1000 + Math.random() * 9000)`
FR-058	CMS Pages	slug	Page content	Admin-managed; rendered at /pages/:slug on frontend

Admin Dashboard Module

ID	Description	Input	Output	Business Logic
FR-059	Dashboard Stats	—	totalOrders, revenue, totalUsers, totalProducts, recentOrders[], salesByDay[]	6 parallel DB queries; 30-day rolling sales chart; only paid orders in revenue
FR-060	Banner CRUD	image (upload or URL), link, position	Banner record	Supports file upload OR URL; is_active toggleable
FR-061	Offer Management	title, discount_pct, target_type, target_id, expires_at	Offer record	Polymorphic target (product/category); shown on product detail page
FR-062	User Management	userId, is_active	Updated user	Cannot modify/delete admin accounts (protected by query WHERE role != 'admin')
FR-063	Notification Send	userId, message	Notification record	In-app; triggered on order status changes; fire-and-forget outside transaction
FR-064	Pickup Status Update	orderId, status	Updated pickup_status	FSM: pending → ready → collected → expired

1.3 Non-Functional Requirements

ID	Category	Requirement	Acceptance Criteria
NFR-001	Security	Password hashing with bcrypt	Min 10 salt rounds; no plaintext passwords in DB or logs
NFR-002	Security	JWT token lifecycle management	Access tokens short-lived; refresh tokens in httpOnly, Secure, SameSite cookies
NFR-003	Security	Content Security Policy	Helmet CSP allowlist: only Razorpay, Google Fonts, and self; no wildcard sources
NFR-004	Security	Auth rate limiting	Max 10 requests/min/IP on all /api/auth/* routes
NFR-005	Security	Global rate limiting	Max 1000 requests/15min/IP across all API routes
NFR-006	Security	Payment signature verification	Every Razorpay payment verified server-side via HMAC-SHA256 before order fulfillment
NFR-007	Security	Admin role enforcement	All /api/admin/* routes require both authGuard (JWT) + adminGuard (role='admin')
NFR-008	Security	SQL injection prevention	100% parameterized queries via pg; no string interpolation in SQL
NFR-009	Performance	Response compression	All API responses gzip-compressed via compression middleware
NFR-010	Performance	DB connection pooling	Single pg.Pool shared across all requests; no per-request connection overhead
NFR-011	Performance	Batch stock pre-check	Stock validation uses single ANY($1) query — not N+1 — before order creation
NFR-012	Performance	Image optimization	All uploads processed by Sharp before storage; appropriate dimensions and quality
NFR-013	Reliability	DB connection resilience	connectDB() retries up to 5 times with exponential backoff (2s, 4s, 6s, 8s, 10s)
NFR-014	Reliability	Atomic order fulfillment	fulfillOrder() uses BEGIN/FOR UPDATE/COMMIT; full ROLLBACK on any failure
NFR-015	Reliability	Process crash prevention	unhandledRejection + uncaughtException handlers log and exit cleanly (code 1)
NFR-016	Scalability	Stateless authentication	JWT-based; no server-side session store; horizontal scaling possible
NFR-017	Scalability	12-factor config	All secrets and URLs via .env; no hardcoded values in source (post-remediation)
NFR-018	Scalability	Health & readiness endpoints	/api/health (liveness) and /api/ready (DB ping) for orchestration probes
NFR-019	Maintainability	Layered architecture	Business logic in controllers/services only; no logic in route handlers or query files
NFR-020	Availability	PWA offline support	Workbox caches all static assets + Google Fonts; app shell available offline
NFR-021	Data Integrity	Non-negative stock guarantee	GREATEST(stock - qty, 0) on deduction; FOR UPDATE row lock prevents race conditions

1.4 Data Requirements

Core Entities

Entity / Table	Key Fields	Relationships	Constraints
`users`	id, full_name, email, phone, password_hash, role, first_citizen_points, is_active	→ orders, addresses, cart_items, wishlists, reviews	email UNIQUE; role ENUM(user/admin); points ≥ 0
`products`	id, title, slug, brand_id, category_id, sub_category_id, base_price, discount_pct, gender, stock, status, is_deal, is_returnable	→ product_variants, product_images, product_attributes; ← brands, categories	slug UNIQUE; status ENUM(active/inactive); stock fallback when no variants
`product_variants`	id, product_id, size, color, sku, stock, extra_price	→ cart_items, order_items, inventory	stock ≥ 0; default variant: size=NULL, color=NULL
`product_images`	id, product_id, url, is_primary, sort_order	← products	One primary per product enforced by app logic
`product_attributes`	id, product_id, label, value, sort_order, section	← products	section ENUM(highlights/specifications)
`categories`	id, name, slug, parent_id	self-referential parent/child	slug UNIQUE; recursive CTE used in product queries
`brands`	id, name, slug, logo_url	→ products	slug UNIQUE
`cart_items`	id, user_id, product_id, variant_id, quantity	← users, product_variants	UNIQUE(user_id, variant_id) for upsert logic
`orders`	id, user_id, address_id, coupon_id, subtotal, discount, shipping, total, points_earned, payment_method, payment_status, status, delivery_type, delivery_method, store_id, pickup_status, pickup_pin, expected_by, razorpay_order_id	← users, addresses; → order_items; ← coupons, stores	status ENUM; payment_status ENUM; delivery_type ENUM
`order_items`	id, order_id, product_id, variant_id, quantity, unit_price	← orders, products, product_variants	unit_price snapshot at purchase time

`addresses`	id, user_id, label, line1, line2, city, state, pincode, full_name, phone, is_deleted	← users	Soft-deleted; is_deleted=true never shown
`coupons`	id, code, discount_type, discount_value, min_order, max_uses, used_count, per_user_limit, expires_at, is_active	→ orders	code UNIQUE; used_count increments atomically
`inventory`	id, variant_id, warehouse_id, stock_quantity, low_stock_threshold	← product_variants, warehouses	NOT auto-created on variant add; manually registered via Inventory module
`inventory_logs`	id, inventory_id, action_type, quantity, notes, admin_id, created_at	← inventory	action_type ENUM(order_deduction/return_restock/manual_adjustment/variant_stock_set)
`warehouses`	id, name, location, manager_name, contact_number, is_active	→ inventory	Soft-deleted via is_active=false
`stores`	id, name, city, state, address, pincode, timing, is_active	→ orders (store_id)	Pickup locations only; separate from warehouses
`banners`	id, image_url, link, position, is_active	standalone	position for display ordering
`offers`	id, title, discount_pct, target_type, target_id, expires_at, is_active	polymorphic target	target_type ENUM(product/category)
`notifications`	id, user_id, message, read_at, created_at	← users	read_at=NULL means unread
`reviews`	id, user_id, product_id, rating, comment, created_at	← users, products	hasPurchasedProduct() guard; rating 1-5
`pages`	id, title, slug, content, is_active	standalone	slug UNIQUE; CMS content
`payment_methods`	id, method, is_active, display config	standalone	Admin-configured; controls storefront payment options
`express_delivery_pincodes`	pincode, city, is_express, delivery_hrs	standalone	PK = pincode (6 digits)

Stock Source-of-Truth Rules (COALESCE Logic)

Scenario	Stock Source	Query Used
Product has variants	SUM(product_variants.stock)	COALESCE((SELECT SUM(pv.stock) FROM product_variants pv WHERE pv.product_id = p.id), p.stock)

Product has no variants	products.stock (fallback)	Same COALESCE — returns p.stock when subquery is NULL
Order fulfillment deduction	product_variants.stock (primary) + inventory.stock_quantity (synced)	FOR UPDATE lock → UPDATE product_variants → UPDATE inventory
Inventory module view	inventory.stock_quantity (display only)	LEFT JOIN — variant shown even without inventory record

1.5 User Roles & Permissions

Role	Module	Read	Write	Delete	Notes
Guest	Products / Search / Categories	✅	❌	❌	Browse-only
Guest	Cart	✅ (local)	✅ (localStorage)	✅ (local)	No DB cart; merged on login
Guest	Orders / Wishlist / Reviews	❌	❌	❌	Must register
Customer	Products / Search / Banners	✅	❌	❌	Storefront only
Customer	Cart (DB)	✅	✅	✅ own items	Authenticated; variant stock checked
Customer	Orders	✅ own	✅ (place)	✅ cancel (pending/confirmed)	Cannot view others' orders
Customer	Addresses	✅ own	✅	✅ (soft)	Soft-delete only
Customer	Reviews	✅	✅ (purchased)	❌	hasPurchasedProduct() gate
Customer	Loyalty Points	✅ own	System-managed	❌	Auto earn/deduct by system
Admin	All Admin Routes	✅	✅	✅	role='admin' enforced by adminGuard
Admin	User Management	✅	✅ (non-admin)	✅ (non-admin)	Cannot modify other admin accounts
Admin	Inventory / Warehouse	✅	✅	✅ (soft)	Full access; logs all changes
Admin	Order Status	✅	✅ (any status)	❌	No hard delete of orders

1.6 API Interface Requirements

Base: http://localhost:5000/api | Envelope: { success, data?, message? }

Auth, Products & Cart

Method	Endpoint	Auth	Description
POST	/auth/register	None	Register user
POST	/auth/login	None	Login (rate-limited)
POST	/auth/refresh	Cookie	Refresh access token
POST	/auth/logout	Bearer	Logout
POST	/auth/send-otp	None	Send OTP via SMS
POST	/auth/verify-otp	None	Verify OTP
GET	/products	None	List with filters/sort/page
GET	/products/:slug	None	Product detail + variants + images + attrs
GET	/products/search?q=	None	Full-text search
GET	/cart	Bearer	View cart
POST	/cart	Bearer	Add item (productId or variantId)
PUT	/cart/:itemId	Bearer	Update quantity
DELETE	/cart/:itemId	Bearer	Remove item
POST	/cart/merge	Bearer	Merge guest cart on login

Payments, Orders & Account

Method	Endpoint	Auth	Description
GET	/payments/methods	None	List active payment methods
POST	/payments/create-order	Bearer	Create Razorpay order + pre-checks
POST	/payments/create-cod-order	Bearer	Place COD order + fulfill
POST	/payments/verify	Bearer	Verify Razorpay payment + fulfill order
GET	/payments/orders/:id	Bearer	Fetch order detail
GET	/orders	Bearer	Order history (paginated)
POST	/orders/:id/cancel	Bearer	Cancel order (pending/confirmed only)
GET	/account/profile	Bearer	User profile + points
GET	/account/addresses	Bearer	List active addresses
POST	/account/addresses	Bearer	Add address
PUT	/account/addresses/:id	Bearer	Update address
DELETE	/account/addresses/:id	Bearer	Soft-delete address
GET	/notifications	Bearer	List user notifications
PUT	/notifications/:id/read	Bearer	Mark notification read

Admin Routes (Bearer + role=admin required)

Method	Endpoint	Description
GET	/admin/dashboard/stats	Dashboard KPIs + 30-day sales chart
GET/POST/PUT/DELETE	/admin/products	Product CRUD (images via Multer)
GET/POST/PUT/DELETE	/admin/products/:id/variants	Variant management
GET/POST/PUT/DELETE	/admin/products/:id/attributes	Product attributes
GET/POST/DELETE	/admin/products/:id/images	Image management
POST	/admin/products/fill-missing-images	Bulk placeholder image assignment
GET/POST/PUT/DELETE	/admin/categories	Category CRUD (hierarchical)
GET/POST/PUT/DELETE	/admin/brands	Brand CRUD (with logo upload)
GET/POST/PUT/DELETE	/admin/banners	Banner CRUD
GET/POST/PUT/DELETE	/admin/coupons	Coupon CRUD
GET/POST/PUT/DELETE	/admin/offers	Offer management
GET	/admin/orders	All orders with filters
PUT	/admin/orders/:id/status	Update order status
PUT	/admin/orders/:id/pickup-status	Update pickup FSM status
GET/PATCH/DELETE	/admin/users	User management
GET/POST/PUT/DELETE	/admin/payment-methods	Payment method config
GET/POST/PATCH/DELETE	/admin/delivery/stores	Store pickup management
GET/POST/PATCH/DELETE	/admin/delivery/pincodes	Express delivery pincodes
GET/POST/PUT/DELETE	/admin/pages	CMS page management

Inventory Routes (Bearer + role=admin required)

Method	Endpoint	Description
GET	/inventory/dashboard/stats	Stock counts by status + warehouse
GET/POST/PUT/DELETE	/inventory/warehouses	Warehouse CRUD
GET/POST/PUT	/inventory/inventory	Inventory record CRUD
POST	/inventory/inventory/bulk-update	Bulk stock quantity update
POST	/inventory/inventory/:id/adjust	Stock delta adjustment with log
PUT	/inventory/variants/:variantId/stock	setVariantStock() — upserts inventory + syncs variant
GET	/inventory/low-stock	Low stock alert list
GET/POST	/inventory/logs	Inventory audit log

1.7 Assumptions & Constraints

ID	Type	Description
A-01	Assumption	Currency is INR exclusively; Razorpay amounts converted to paise (×100) via Math.round()
A-02	Assumption	Phone numbers with +91 prefix route to Fast2SMS; all others to Twilio
A-03	Assumption	Products with no variants use products.stock as fallback; COALESCE handles both cases
A-04	Assumption	Loyalty points never expire; permanent until redeemed or order cancelled
A-05	Assumption	A review can only be submitted after order.status = 'delivered'
A-06	Assumption	Store pickup and warehouses are separate concepts (stores table ≠ warehouses table)
A-07	Assumption	Image uploads are processed by Sharp and stored on local disk /uploads (no cloud storage)
C-01	Constraint	Inventory row is NOT auto-created when a variant is added via the Products tab — admin must manually register it in the Inventory module
C-02	Constraint	Guest cart relies on client-side localStorage only — no server-side stock validation until merge on login
C-03	Constraint	Bulk stock update (bulkUpdateInventoryStock) has no transaction wrapper — partial failure leaves inconsistent state
C-04	Constraint	No refresh token revocation store — logout does not truly invalidate tokens until natural expiry
C-05	Constraint	Local file storage blocks horizontal scaling — S3/CDN migration required for multi-server deployment
C-06	Constraint	No API versioning (/api/v1/) — breaking changes affect all clients immediately
C-07	Constraint	No automated test suite — all testing is manual; no CI/CD pipeline configured
C-08	Constraint	Migration scripts are numbered manually (migrate.js, migrate-phase5.js…) with no migration runner
C-09	Constraint	Razorpay only; no PayPal, UPI QR, or digital wallet support
C-10	Constraint	Store pickup PIN is random (not cryptographically secure); adequate for low-stakes use but not banking-grade

1.8 Glossary

Term	Definition
SKU	Stock Keeping Unit — unique identifier per product variant (stored in product_variants.sku)
COALESCE Stock	SQL pattern used in product queries: COALESCE((SELECT SUM(variant.stock)...), product.stock) — returns variant sum if variants exist, otherwise falls back to product-level stock
First Citizen Points	Loyalty reward program; customers earn 1 point per ₹100 spent; redeemable at checkout (1 pt = ₹1); capped at 20% of post-coupon subtotal per order
setVariantStock()	Server function in inventory.js that upserts an inventory record for a variant AND updates product_variants.stock simultaneously — the canonical way to set stock via the Inventory module
fulfillOrder()	Atomic DB transaction function in orders.js that: inserts order, inserts items, locks variants (FOR UPDATE), validates stock, deducts stock, syncs inventory, awards points, marks coupon used, clears cart — all in one BEGIN/COMMIT block
authGuard	Express middleware that validates JWT Bearer token and attaches req.user; applied to all protected routes
adminGuard	Express middleware that enforces req.user.role === 'admin'; applied after authGuard on all /api/admin/* and /api/inventory/* routes
lineTotal	Cart item subtotal computed as unitPrice × quantity in the DB cart query; returned pre-computed in cart response
paise	Indian sub-unit of currency; 1 INR = 100 paise; Razorpay API requires all amounts in paise (Math.round(amount × 100))
Pickup PIN	4-digit random code generated at store-pickup order creation for collection
Soft Delete	Setting is_deleted=true (addresses) or is_active=false (warehouses) instead of removing the DB row
Default Variant	A product_variants row with size=NULL and color=NULL, auto-created by cart.controller.js when a product without explicit variants is added to cart
buildResponse()	Cart controller helper that computes subtotal and itemCount from raw cart items array; ensures consistent cart response shape
calcShipping()	Shared function in payments.controller.js; determines shipping cost from deliveryType and subtotal; single source of truth for shipping logic

PART 2

Architecture & Code Review

Independent assessment of code quality, security posture, performance characteristics, and architectural integrity. Findings reference actual function names, file paths, and SQL patterns found in the source code.

2.1 Architecture Assessment

Dimension	Finding	Status
Pattern	Layered Monolith — React SPA + Express REST API. Clean separation of concerns.	✅ **Appropriate**
Layer Separation	Routes → Controllers → Queries/Services strictly followed. No business logic in route handlers.	✅ **Clean**
Admin Module	admin.routes.js is 566 lines with inline route handlers — no dedicated admin controllers. Untestable.	⚠️ **God File**
Coupling	Controllers are loosely coupled via function imports. DB pool is a shared singleton — acceptable.	✅ **Acceptable**
Reusability	calcShipping(), buildResponse(), getProductPlaceholder() are good shared helpers. More extraction needed.	⚠️ **Partial**
Config	All secrets via .env — 12-factor compliant. Health + readiness endpoints present.	✅ **Good**
Frontend Architecture	Zustand stores per concern (auth, cart, wishlist, UI). Lazy-loaded routes with Suspense. PWA-ready.	✅ **Well structured**

2.2 Code Quality Review

Area	Observation	Status
Naming Conventions	Consistent camelCase throughout; descriptive function names (fulfillOrder, setVariantStock, buildResponse).	✅ **Good**
DRY Violations	Stock pre-check loop was duplicated in createOrder + createCodOrder (fixed). calcShipping defined once — good.	⚠️ **Mostly fixed**
Function Complexity	CheckoutPage.jsx ~889 lines; fulfillOrder() ~100 lines but justified as a transaction unit.	⚠️ **CheckoutPage needs splitting**
Dead Code	Old createOrder() / addOrderItems() / deductStock() / awardPoints() / markCouponUsed() functions in orders.js are superseded by fulfillOrder() but still exported. Unused.	❌ **Remove dead exports**
Comments	Meaningful inline comments on complex logic (COALESCE, recursive CTE, FOR UPDATE). Not over-commented.	✅ **Good**
Console Logs	authGuard.js debug logs removed. Some [createOrder] logs remain — acceptable for dev, gate behind DEBUG env var for prod.	⚠️ **Minor**
Joi Usage	Joi installed but not used anywhere — no request body validation layer exists.	❌ **Critical gap**
Magic Strings	'standard', 'express', 'store_pickup' delivery type strings used in 5+ places without constants.	⚠️ **Extract constants**

2.3 Security Audit

Check	Finding	Status
Password Hashing	bcrypt with default rounds. No plaintext anywhere.	✅ **Secure**
JWT Implementation	Short-lived access tokens; refresh in httpOnly cookie. No revocation store.	⚠️ **Logout gap**
SQL Injection	100% parameterized queries. No string interpolation in SQL. Safe.	✅ **Secure**
Admin Authorization	authGuard + adminGuard applied at router level (router.use(authGuard, adminGuard)). All admin routes covered.	✅ **Secure**
Payment Security	HMAC-SHA256 signature verification server-side before any order fulfillment. Correct.	✅ **Secure**
Exposed Secrets (Historical)	Twilio SID + Fast2SMS key were in SMS-NOTIFICATION-GUIDE.md (commit a29ee3f). Scrubbed from history. CREDENTIALS MUST BE ROTATED.	❌ **Rotate immediately**
CSP Headers	Helmet CSP configured with explicit allowlist — Razorpay + Google Fonts only.	✅ **Secure**
CORS	Allows CLIENT_URL + any localhost port in dev. Strict in production.	✅ **Acceptable**
Input Validation	No Joi schemas on any endpoint. Only ad-hoc checks (if !addressId). XSS risk from unvalidated string inputs.	❌ **Add validation**
Address Ownership	Validated via array scan (addresses.find()) not DB query — inefficient but functionally correct.	⚠️ **Use DB WHERE user_id=$1 AND id=$2**
Rate Limiting	Auth: 10/min; Global: 1000/15min. In-memory store — resets on restart; Redis recommended for prod.	⚠️ **Use Redis for prod**

2.4 Performance Analysis

Issue	Location	Impact	Status
N+1 Stock Pre-check (fixed)	payments.controller.js	Was: 1 query per cart item. Now: single ANY($1) batch query.	✅ **Fixed**
Stock loop in fulfillOrder()	orders.js fulfillOrder()	Still loops per item for FOR UPDATE lock — intentional (cannot batch row locks)	✅ **Correct**
RANDOM() sort	products.js getProducts()	ORDER BY RANDOM() is O(n log n) full table scan — disables index use	⚠️ **Cache or precompute**
No query indexes documented	All migration files	Missing indexes on: orders.user_id, cart_items.user_id, products.slug, product_variants.product_id	❌ **Add indexes**
Razorpay singleton	payments.service.js getRazorpay()	New Razorpay instance created per call. Should be module-level singleton.	⚠️ **Minor overhead**
No caching	Public endpoints	Categories, banners, homepage data queried from DB on every request — cacheable.	⚠️ **Add Redis cache**
Parallel queries	getProductBySlug(), adminGetDashboardStats()	Promise.all() used correctly for independent queries.	✅ **Good**

2.5 Critical Gaps Found

#	Gap	Location	Risk	Recommendation
1	Inventory row NOT auto-created on variant add	admin.routes.js POST /products/:id/variants	HIGH	Add createInventory() call after addVariant() or add a DB trigger on product_variants INSERT
2	Guest cart has no server-side stock check	localStorage (client only)	HIGH	Cart merge endpoint (POST /cart/merge) should validate stock per item before upserting
3	Bulk stock update has no transaction	inventory.js bulkUpdateInventoryStock()	HIGH	Wrap in BEGIN/COMMIT; partial failure currently leaves inconsistent state
4	No Joi validation on any request body	All controllers	HIGH	Add Joi schemas — library already installed; implement on checkout, auth, and admin routes first
5	Dead code in orders.js	orders.js lines 4-50	MED	Remove createOrder(), addOrderItems(), deductStock(), awardPoints(), markCouponUsed() — all superseded by fulfillOrder()
6	Zero test coverage	Entire codebase	HIGH	Start with: verifySignature(), validateCoupon(), calcShipping(), fulfillOrder() integration test
7	No missing index definitions	Migration files	MED	Add: CREATE INDEX ON orders(user_id); CREATE INDEX ON cart_items(user_id); CREATE UNIQUE INDEX ON products(slug);

2.6 Review Scorecard

Area	Score (1–10)	Status	Priority
Architecture	7.5 / 10	✅ **Good**	LOW
Code Quality	6.5 / 10	⚠️ **Acceptable**	MEDIUM
Security	6.0 / 10	⚠️ **Needs work**	HIGH
Performance	6.0 / 10	⚠️ **Missing indexes/cache**	MEDIUM
Data Integrity	7.0 / 10	✅ **Good (transaction fixed)**	MEDIUM
Error Handling	7.0 / 10	✅ **Central handler + process guards**	LOW
Testing	1.0 / 10	❌ **Zero tests**	CRITICAL
Overall	**5.9 / 10**	**Pre-production MVP**	**Not production-ready**

2.7 Top 5 Priority Fixes Before Production

#	Issue	Fix	Effort
1	🔴 **Rotate exposed credentials**	Regenerate Twilio SID/auth token and Fast2SMS API key on their dashboards immediately. Update .env. The old credentials were in git commit a29ee3f and may have been scraped.	30 min
2	🔴 **Auto-create inventory on variant add**	In admin.routes.js POST /products/:id/variants, after addVariant(), call createInventory({ variant_id: variant.id, stock_quantity: variant.stock }). This closes the gap between the Products tab and Inventory module.	1 hour
3	🔴 **Add Joi validation to all endpoints**	Joi is installed. Add schemas for: register, login, createOrder, createCodOrder, addToCart, updateQty. Prevents invalid data reaching DB and closes XSS surface.	1 day
4	🔴 **Wrap bulk stock update in transaction**	In inventory.js bulkUpdateInventoryStock(), wrap all updates in a single client.query('BEGIN') / COMMIT block. If any update fails, ROLLBACK to maintain consistency.	2 hours
5	🔴 **Add critical DB indexes**	Run migration: CREATE INDEX ON orders(user_id); CREATE INDEX ON cart_items(user_id); CREATE INDEX ON product_variants(product_id); CREATE UNIQUE INDEX ON products(slug). Without these, queries degrade as data grows.	1 hour

PART 3

Complete E-Commerce Workflow Documentation

End-to-end workflow covering all 10 operational phases for Admin, Customer, and Guest actors. Includes phase triggers, API calls, DB tables affected, business rules, and outcomes.

3.1 System Workflow Overview

Admin Actor

The admin accesses the platform at `/admin` and logs in using credentials verified against the users table (role='admin'). Once authenticated, the admin can manage the full product catalogue — creating products, adding size/color variants with stock levels, uploading images, and optionally registering those variants in the Inventory module (linked to a physical warehouse). The admin configures homepage banners, promotional offers, discount coupons, delivery options (express pincodes, store pickup locations), and CMS pages. When orders arrive, the admin reviews them on the Orders dashboard, updates status through the lifecycle (confirmed → shipped → delivered), and monitors inventory health via the Inventory Dashboard with low-stock alerts and audit logs.

Logged-in Customer Actor

A registered customer authenticates via email/password or OTP. On login, any items in their localStorage guest cart are automatically merged into their DB cart. They browse products using category navigation, filters (price, brand, gender, discount), full-text search, and sorting. On the product detail page they select a size/color variant, check stock availability, and add to their DB cart. At checkout, they select a saved address (or add a new one), choose a delivery option (Standard, Express subject to pincode availability, or Store Pickup), optionally apply a coupon code, optionally redeem First Citizen loyalty points (up to 20% of subtotal), and proceed to pay via Razorpay (online) or COD. After payment, the order is atomically fulfilled, their cart is cleared, points are awarded, and an SMS confirmation is sent. They can track orders, view history, cancel eligible orders, and leave reviews on delivered products.

Guest Customer Actor

A guest can browse the full product catalogue, view product details including pricing and availability, and add items to a local cart stored in their browser's localStorage. The app reads variant stock from the API for display purposes. No server-side stock reservation happens for guests. When the guest registers or logs in, their localStorage cart is sent to the POST /cart/merge endpoint, which upserts each item into the authenticated DB cart. From that point they follow the customer checkout flow.

3.2 Phase-by-Phase Workflow

PHASE 1 — Product Creation (Admin)

Actor:	Admin
Trigger:	Admin clicks "Add Product" in Admin Panel → Products
Steps:	1. Fill product form (title, brand, category, price, discount%, gender, description, status) 2. Upload product images (Multer receives files; Sharp resizes/optimizes; saved to /uploads) 3. Submit → POST /admin/products
API Called:	POST /admin/products (multipart/form-data with images)
DB Tables Affected:	products (INSERT), product_images (INSERT per file, first=primary)
Business Rules:	slug must be unique; status defaults to 'active'; is_deal and is_returnable default to false/true
Success Outcome:	Product visible in admin list; if status='active' immediately visible on storefront
Failure Outcome:	Duplicate slug → DB unique constraint error → 500 (should be caught + 400)

PHASE 2 — Variant Addition & Stock Setup

Actor:	Admin
Trigger:	Admin opens product → Variants tab → "Add Variant"
Steps:	1. Enter size, color, SKU, stock quantity, extra_price 2. POST /admin/products/:id/variants 3. Variant appears in variant list with stock count ⚠️ NOTE: No inventory record is auto-created at this step
API Called:	POST /admin/products/:id/variants
DB Tables Affected:	product_variants (INSERT)
Business Rules:	stock ≥ 0; extra_price added to base_price for final variant price; variant stock overrides products.stock in COALESCE queries
Success Outcome:	Variant selectable on product detail page; stock shown to customers
Failure Outcome:	Variant without inventory record — Inventory module shows it but cannot track warehouse location until Step 3

PHASE 3 — Inventory Module Registration

Actor:	Admin
Trigger:	Admin navigates to Inventory module → assigns variant to warehouse
Steps:	1. Create warehouse if needed: POST /inventory/warehouses 2. Register variant: POST /inventory/inventory {variant_id, warehouse_id, stock_quantity} OR use the faster: PUT /inventory/variants/:variantId/stock 3. setVariantStock() upserts inventory row + updates product_variants.stock + creates inventory log
API Called:	POST /inventory/warehouses; PUT /inventory/variants/:variantId/stock
DB Tables Affected:	warehouses (INSERT), inventory (UPSERT), product_variants (UPDATE stock), inventory_logs (INSERT with action='variant_stock_set')
Business Rules:	setVariantStock() is the authoritative function — it keeps both tables in sync; inventory.stock_quantity mirrors product_variants.stock after this call
Success Outcome:	Variant appears in Inventory Dashboard; low-stock alerts now active; warehouse association visible
Failure Outcome:	If skipped: variant tracked only in product_variants; inventory module shows it with no warehouse; low-stock alerts inactive for this variant

PHASE 4 — Customer Browses Storefront

Actor:	Guest or Customer
Trigger:	User opens app at http://localhost:5173
Steps:	1. Homepage loads → parallel API calls: banners, featured products, categories, offers 2. User navigates category (e.g., /women) → GET /products?gender=women&category=women 3. Recursive CTE resolves parent category → all subcategory products included 4. User applies filters (price, brand, discount) → query params updated → new API call 5. User clicks product → GET /products/:slug → images + variants + attributes loaded 6. Stock shown per variant via COALESCE logic
API Called:	GET /home; GET /products; GET /products/:slug; GET /products/search
DB Tables Affected:	Read-only: products, product_variants, product_images, product_attributes, brands, categories, banners, offers
Business Rules:	Only status='active' products shown; COALESCE stock: SUM(variant.stock) if variants exist, else products.stock; offers displayed if not expired
Success Outcome:	Product catalogue rendered with live stock and pricing
Failure Outcome:	DB unavailable → /api/ready returns 503; frontend shows error state

PHASE 5 — Add to Cart (Logged-in + Guest)

Actor:	Customer (DB cart) or Guest (localStorage)
Trigger:	User clicks "Add to Bag" on product detail page after selecting variant
Steps (Customer):	1. POST /cart {variantId, quantity} 2. If no variantId: server finds/creates default variant (size=NULL, color=NULL) 3. Server checks product_variants.stock > 0 → 409 if out of stock 4. addOrUpdateItem(): INSERT ... ON CONFLICT (user_id, variant_id) DO UPDATE SET quantity = quantity + $qty 5. Returns updated cart with subtotal and itemCount
Steps (Guest):	1. Item added to localStorage array (no server call) 2. Stock display is from previous product detail API call (not real-time) 3. On login: POST /cart/merge {items:[]} sends localStorage cart to server ⚠️ No stock validation at merge time — gap exists
API Called:	POST /cart; POST /cart/merge (on login)
DB Tables Affected:	cart_items (UPSERT), product_variants (READ stock check only)
Business Rules:	Stock must be > 0 to add (authenticated users); UPSERT increments quantity if item exists; quantity must be ≥ 1
Success Outcome:	Item in cart; subtotal updated; cart badge in navbar increments
Failure Outcome:	Out of stock → 409; item not found → 404

PHASE 6 — Checkout & Payment (Razorpay + COD)

Actor:	Customer
Trigger:	Customer clicks "Proceed to Checkout" from cart
Steps (Razorpay):	1. Step 1 (Address): Select/add delivery address 2. Step 2 (Delivery): Choose Standard / Express (pincode check) / Store Pickup 3. Step 3 (Payment): Enter coupon code → validated server-side; toggle Use Points; select Razorpay 4. POST /payments/create-order → batch stock check (ANY query) → coupon validate → points cap 20% → calcShipping() → Razorpay API creates order 5. Response: razorpayOrderId + amount + keyId sent to frontend 6. Razorpay modal opens → customer pays 7. On success: POST /payments/verify {razorpayOrderId, paymentId, signature, ...totals} 8. Server: verifySignature() → HMAC match → fulfillOrder() → redirect to /order-success
Steps (COD):	1–3 same as above 4. POST /payments/create-cod-order → same pre-checks → direct fulfillOrder() call 5. payment_status='pending' (paid only on delivery)
API Called:	GET /stores; GET /account/addresses; POST /payments/create-order; POST /payments/verify; POST /payments/create-cod-order

DB Tables Affected:	orders (INSERT), order_items (INSERT), product_variants (UPDATE stock), inventory (UPDATE), cart_items (DELETE), users (UPDATE points), coupons (UPDATE used_count)
Business Rules:	Shipping: Standard ≥₹999 free else ₹99; Express ₹199; Pickup ₹0. Points: MIN(balance, FLOOR((subtotal-discount)×0.20)). Coupon: validated against active/expired/min_order/per_user/global limits.
Success Outcome:	Order created; cart cleared; SMS sent; in-app notification fired; redirect to order success page
Failure Outcome:	Out of stock → 409; invalid signature → 400; Razorpay error → 502 with message; coupon invalid → 400

PHASE 7 — Order Fulfillment & Stock Deduction (atomic)

Actor:	System (fulfillOrder() transaction)
Trigger:	Called from verifyPayment() or createCodOrder() after all pre-checks pass
Steps:	1. client.query('BEGIN') 2. INSERT INTO orders (...) RETURNING * → order record created 3. INSERT INTO order_items (batch values template) → all items inserted 4. FOR EACH item with variantId: a. SELECT stock FROM product_variants WHERE id=$1 FOR UPDATE (row lock) b. IF stock < quantity → throw 409 error → ROLLBACK c. UPDATE product_variants SET stock = stock - $qty WHERE id=$variantId d. UPDATE inventory SET stock_quantity = GREATEST(stock_quantity - $qty, 0) WHERE variant_id=$variantId e. INSERT INTO inventory_logs (action='order_deduction', qty=-qty) 5. IF pointsEarned > 0: UPDATE users SET first_citizen_points = first_citizen_points + $pts 6. IF couponId: UPDATE coupons SET used_count = used_count + 1 7. DELETE FROM cart_items WHERE user_id=$userId 8. client.query('COMMIT') 9. (Outside transaction): createNotification() fire-and-forget
DB Tables Affected:	orders, order_items, product_variants, inventory, inventory_logs, users, coupons, cart_items, notifications
Business Rules:	FOR UPDATE prevents race condition on last item. GREATEST prevents negative stock. pointsEarned = FLOOR(total/100). All steps atomic — any failure rolls back completely.
Success Outcome:	Order confirmed; stock decremented in both product_variants and inventory; points awarded; coupon counted; cart empty
Failure Outcome:	Any error → ROLLBACK → no partial state; customer sees error message; Razorpay payment captured but order not created → requires manual reconciliation (gap)

PHASE 8 — Order Cancellation & Stock Restoration

Actor:	Customer or Admin
Trigger:	Customer clicks "Cancel Order" on order detail page; or Admin changes order status
Steps:	1. POST /orders/:id/cancel (customer) or PUT /admin/orders/:id/status {status:'cancelled'} 2. client.query('BEGIN') 3. UPDATE orders SET status='cancelled' WHERE id=$1 AND user_id=$2 AND status IN ('pending','confirmed') RETURNING * 4. IF no rows returned → ROLLBACK (order not in cancellable state) 5. SELECT variant_id, quantity FROM order_items WHERE order_id=$1 6. FOR EACH item: a. UPDATE product_variants SET stock = stock + $qty WHERE id=$variantId b. UPDATE inventory SET stock_quantity = stock_quantity + $qty WHERE variant_id=$variantId RETURNING id c. INSERT INTO inventory_logs (action='return_restock', qty=+qty, notes='Order #X cancelled') 7. IF points_earned > 0: UPDATE users SET first_citizen_points = GREATEST(pts - earned, 0) 8. client.query('COMMIT')
API Called:	POST /orders/:id/cancel
DB Tables Affected:	orders, product_variants, inventory, inventory_logs, users
Business Rules:	Only pending/confirmed orders cancellable. Stock fully restored. Points from this order deducted (GREATEST prevents negative). Coupon used_count NOT decremented (gap).
Success Outcome:	Order cancelled; stock restored; points adjusted; customer can re-order
Failure Outcome:	Order already shipped/delivered → null returned → 404 "Order not found or cannot be cancelled"

PHASE 9 — Storefront Visibility & Stock Filters

Actor:	System (query layer)
Trigger:	Any product listing or detail API call
Steps:	1. getProducts() builds WHERE p.status = 'active' as base condition 2. If category filter: recursive CTE resolves all child category IDs 3. Stock computed: COALESCE((SELECT SUM(pv.stock) FROM product_variants pv WHERE pv.product_id = p.id), p.stock) 4. Returns stock alongside each product — frontend uses this to show "In Stock" / "Out of Stock" 5. Variant-level stock shown on product detail page (SELECT * FROM product_variants WHERE product_id=$1) 6. Admin soft-delete sets status='inactive' → product immediately disappears from all storefront queries
API Called:	GET /products; GET /products/:slug
DB Tables Affected:	Read-only: products, product_variants, categories (recursive CTE)
Business Rules:	COALESCE: if variants exist → SUM(variant.stock); else → products.stock. Frontend filters: stock=0 → "Out of Stock" badge, add-to-cart disabled. Categories: browsing /category/men shows all men + subcategory products via recursive CTE.
Success Outcome:	Accurate real-time stock visibility; category hierarchy browsing works correctly
Failure Outcome:	If inventory row not created but variant exists: stock from product_variants — functionally correct, inventory module just has no warehouse mapping

PHASE 10 — Loyalty Points & Coupon Lifecycle

Actor:	System + Customer
Trigger:	Customer places order / cancels order / applies coupon at checkout
Points Earn:	pointsEarned = FLOOR(total / 100) → awarded inside fulfillOrder() transaction atomically
Points Redeem:	1. Customer toggles "Use Points" at checkout 2. Server fetches user.first_citizen_points 3. maxDiscount = FLOOR((subtotal - couponDiscount) × 0.20) 4. pointsDiscount = MIN(userPoints, maxDiscount) 5. Deducted from order total; separate deduction via UPDATE users after verify (gap: not inside fulfillOrder transaction)
Points on Cancel:	Earned points reversed: UPDATE users SET first_citizen_points = GREATEST(pts - earned, 0) inside cancel transaction
Coupon Flow:	1. Customer enters code → POST /coupons/validate (or inline in create-order) 2. validateCoupon(): active? → not expired? → min_order met? → per_user_limit? → global max_uses? → compute discount 3. Coupon ID stored in order; used_count incremented inside fulfillOrder() transaction ⚠️ On cancel: used_count NOT decremented — gap
DB Tables Affected:	users (points UPDATE), coupons (used_count UPDATE), orders (discount, coupon_id)
Business Rules:	Points cap: 20% of post-coupon subtotal. 1 point = ₹1. No expiry on points. Coupon: flat or percentage discount; per-user and global usage limits enforced separately.
Success Outcome:	Points correctly earned, capped, and redeemed. Coupon validated and usage tracked.
Failure Outcome:	Coupon expired/overused → 400 with reason message. Points balance never goes negative (GREATEST guard).

3.3 Stock Management Decision Tree

```
STOCK SOURCE OF TRUTH DECISION TREE
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Q1: Does the product have any rows in product_variants?
│
├─ NO  → Stock = products.stock (fallback field)
│        Shown as: COALESCE(NULL, products.stock) = products.stock
│        Edit via: PUT /admin/products/:id {stock: N}
│
└─ YES → Stock = SUM(product_variants.stock) for all variants
         Shown as: COALESCE(SUM(pv.stock), products.stock)
         products.stock is IGNORED when variants exist

   Q2: Is this variant registered in the Inventory module?
   │
   ├─ NO  → Stock tracked in product_variants only
   │        No warehouse mapping; no low-stock alert
   │        Updated via: PUT /admin/products/variants/:id
   │
   └─ YES → Stock synced across BOTH tables:
            • product_variants.stock (source of truth for orders)
            • inventory.stock_quantity (mirrors; used for dashboard)
            Updated via: PUT /inventory/variants/:variantId/stock
                         → setVariantStock() keeps both in sync

STOCK STATE TABLE
═══════════════════════════════════════════════════════════════════
State                                | Storefront   | Inventory Module
─────────────────────────────────────┼──────────────┼─────────────────
No variants, products.stock = 0      | Out of Stock | Not tracked
No variants, products.stock > 0      | In Stock     | Not tracked
Variants exist, all stock = 0        | Out of Stock | Depends on reg.
Variants exist, some stock > 0       | In Stock     | Depends on reg.
Variant registered, inv. row exists  | In Stock     | Tracked + alerts
Order placed → stock decremented     | Real-time    | Synced + logged
Order cancelled → stock restored     | Real-time    | Synced + logged
═══════════════════════════════════════════════════════════════════
```

3.4 Transaction Safety Map

Operation	Transaction?	Row Lock?	Tables Modified	Rollback Condition
Order Placement (fulfillOrder)	✅ BEGIN/COMMIT	✅ FOR UPDATE on variants	orders, order_items, product_variants, inventory, inventory_logs, users, coupons, cart_items	stock < quantity ordered; any DB error
Order Cancellation (cancelOrder)	✅ BEGIN/COMMIT	❌ No row lock	orders, product_variants, inventory, inventory_logs, users	Order not in pending/confirmed status
Bulk Stock Update	❌ No transaction	❌ No row lock	inventory, product_variants, inventory_logs	None — partial failure possible
setVariantStock()	⚠️ Single client block	❌ No explicit lock	product_variants, inventory (UPSERT), inventory_logs	Variant not found
adjustStock()	⚠️ Single client block	❌ No explicit lock	inventory, product_variants, inventory_logs	Inventory item not found
Cart Add/Update	❌ Single query	❌ No lock	cart_items	Stock check is pre-flight only; no atomic reserve
Coupon Validation + Apply	✅ Inside fulfillOrder	❌ No lock on coupon row	coupons (used_count)	TOCTOU risk under high concurrency on last use
Points Redemption Deduction	❌ Outside fulfillOrder	❌ No lock	users	None — race condition possible under concurrency

3.5 Data Flow Diagrams

Product Creation Flow

Order Placement Flow (Razorpay)

```
              ├─ INSERT order_items → line items
              ├─ FOR EACH variant: FOR UPDATE → validate → UPDATE stock
              ├─ UPDATE inventory (sync)
              ├─ INSERT inventory_logs
              ├─ UPDATE users (points)
              ├─ UPDATE coupons (used_count)
              └─ DELETE cart_items
       [COMMIT]
              |
              ▼
       [SMS: sendOrderConfirmation()] ← fire-and-forget
       [Notification: createNotification()] ← fire-and-forget
              |
              ▼
[Response: { success: true, data: { orderId } }]
       |
       ▼
[Customer: redirect to /order-success/:orderId]
```

Stock Sync Flow

```
Admin sets stock via Inventory Module:
[PUT /inventory/variants/:variantId/stock { stock_quantity: 50 }]
       |
       ▼
[inventory.controller.js: updateVariantStock()]
       |
       ▼
[setVariantStock(variantId, 50, threshold, adminId, notes)]
       |
       ├─ UPDATE product_variants SET stock = 50 WHERE id = $variantId
       |
       ├─ INSERT INTO inventory (variant_id, stock_quantity=50)
       |   ON CONFLICT (variant_id) DO UPDATE SET stock_quantity = 50
       |
       └─ INSERT INTO inventory_logs (action='variant_stock_set', qty=50)
              |
              ▼
[Both product_variants.stock AND inventory.stock_quantity = 50]

When order is placed (inside fulfillOrder transaction):
[UPDATE product_variants SET stock = stock - qty WHERE id = $variantId]
[UPDATE inventory SET stock_quantity = GREATEST(stock_quantity - qty, 0) WHERE variant_id = $variantId]
[INSERT INTO inventory_logs (action='order_deduction', qty=-qty)]

When order is cancelled (inside cancelOrder transaction):
[UPDATE product_variants SET stock = stock + qty WHERE id = $variantId]
[UPDATE inventory SET stock_quantity = stock_quantity + qty WHERE variant_id = $variantId]
[INSERT INTO inventory_logs (action='return_restock', qty=+qty)]
```

EXECUTIVE SUMMARY

ShopperHub — System Maturity Assessment

One-page summary for stakeholders and engineering leadership.

What the System Does

ShopperHub is a complete fashion e-commerce platform with a React SPA frontend, Node.js/Express backend, and PostgreSQL database. It supports the full retail lifecycle: product catalogue management with hierarchical categories and variants, multi-step checkout with Razorpay online payment and COD, atomic order fulfillment with inventory synchronization, loyalty points, coupon engine, SMS notifications, and a comprehensive admin panel covering products, inventory, orders, banners, CMS pages, delivery configuration, and warehouse management.

✅ Top 3 Strengths

- → Atomic order fulfillment: fulfillOrder() uses DB transactions with FOR UPDATE row locks — prevents overselling and data corruption under concurrent orders
- → Layered architecture: strict Routes → Controllers → Services → Queries separation makes the codebase navigable and extensible
- → Security fundamentals: Helmet CSP, JWT in httpOnly cookies, bcrypt passwords, HMAC payment verification, admin role guard — solid baseline

❌ Top 3 Risks

- → Zero test coverage: no unit, integration, or E2E tests for any business logic — fulfillOrder(), verifySignature(), validateCoupon() are all untested. A regression in payment flow could go undetected.
- → Inventory gap: variant stock is NOT automatically registered in the Inventory module when a variant is added — admin must do this manually. Missed setup means warehouse tracking and low-stock alerts are silently disabled.
- → Exposed credentials in git history: Twilio SID + Fast2SMS API key were committed and need immediate rotation — anyone with repo access (including historical clones) has these credentials.

💡 Recommended Next Steps

- → Immediate (Week 1): Rotate all exposed API keys. Auto-create inventory row on variant add. Add Joi validation to checkout and auth endpoints.
- → Short-term (Month 1): Write tests for payment verification, order fulfillment, coupon validation. Add DB indexes (orders.user_id, cart_items.user_id, products.slug). Wrap bulk stock update in transaction.
- → Pre-production (Month 2): Migrate image storage to S3/Cloudflare R2. Add Redis for rate limiting and public endpoint caching. Add API versioning (/api/v1/). Set up CI/CD pipeline with test gate.

📊 Current Maturity: 5.9/10

- → Architecture: 7.5/10 — Clean, well-structured monolith
- → Security: 6.0/10 — Good foundations; missing input validation
- → Data Integrity: 7.0/10 — Transactions present; bulk update gap
- → Testing: 1.0/10 — No automated tests; critical blocker for production
- → Performance: 6.0/10 — No indexes documented; no caching layer
- → **Verdict: Strong MVP foundation. Not production-ready. Estimated 4–6 weeks of hardening required before launch.**